1. True or False.
This question is related to the example using a payroll thread and an ATM withdrawal thread. If the two threads execute concurrently in an interleaved way (i.e., instructions from each would execute alternatively), the integrity of the shared variable B (balance) may get violated.
True

2. In order to address the data-shared problem, we can use
Mutual exclusion

3. If a solution to Mutual Exclusion meets the progress requirement, this means that
a thread NOT in its critical region cannot block another thread from entering its respective critical region

4. To be a critical region,
at least one thread have some instructions writing some shared data while another may be writing or reading the same shared data
at least two threads have some instructions accessing shared data

5. The solution providing mutual exclusion below meets these requirements:

```
Initialization: Turn = 1

while (1){                        while (1){
      while (turn != 1);                while (turn == 0);
      Critical Section                  Critical Section
      turn = 0;                         turn = 1;
      Remainder Section                 Remainder Section
}                                 }
```

None of these answers

6. A busy waiting solution means that ___________.
the entry code uses the CPU to wait before access to the critical region

7. Select the best option to complete code for the second attempt to provide mutual exclusion.

```
Initialization: Interested0 = Interested1 = false

while (1){                        while (1){
      Interested1 = true;
      while (                     [ Select ]
[ Select ]
);                                      while (Interested1
      Critical Section            );
                                        Critical Section
[ Select ]
                                  [ Select ]
      Remainder Section
}                                       Remainder Section
                                  }
```

Interested0
Interested1 = false
Interested0 = true
Interested1
Interested0 = false

8. Select the best option to complete code for the second attempt to provide mutual exclusion. If possible, we want to guarantee that the thread on the left will enter first to the critical region. If impossible, choose for all choices "No guarantee".

```
Initialization: Interested0 = false
; Interested1 = false
```

```
while (1){
      Interested0 = true;
      while (Interested1);
      Critical Section
      Interested0 = false
      Remainder Section
}
```

```
while (1){
      Interested1 = true;
      while (Interested0);
      Critical Section
      Interested1 = false
      Remainder Section
}
```

?
?

9. True or false.
If the CPU does not offer the Test-and-set instruction, then just implement it using assembly programming.
False

10. Check the busy waiting solutions providing mutual exclusion.
Peterson's Solution
Spinlocks
Baker's algorithm
Using test-and-set instruction

11. If using semaphores, which instructions would you use in the entry code of a critical region?
wait(S)
down(S)

12. A busy waiting semaphore is called a ___________.
spinlock

13. A key property that a semaphore must have (to help with the mutual exclusion) is:
atomicity

14. A process calls the instruction wait(S) with S = 3. After the call to wait(S), value of S will be _____.
2

15. A process calls the instruction wait(S) with S = 1. After the call to wait(S), the state of the processes will likely be ______.
running

16. A key property that Test-and-set swap instructions must have (to help with the mutual exclusion) is:
atomicity